S.No: 3	Exp. Name: ***Simple Calculator***	**Date: 2026-08-28**

Aim:

You are provided with an HTML file that defines the structure of a basic calculator. The calculator supports the following arithmetic operations:

- Addition (+)
- Subtraction (-)
- Multiplication (*)
- Division (/)

Your task is to complete the JavaScript file (script.js) so that the calculator functions correctly.

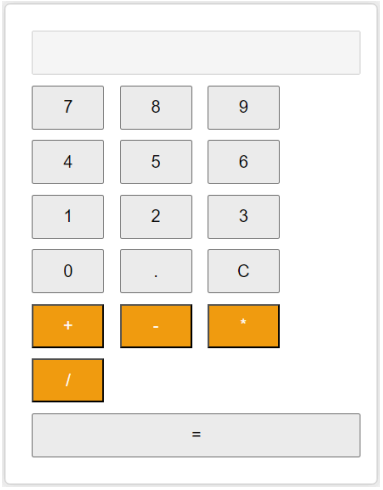
Provided Files

HTML File (index.html)

The HTML file already includes:

- A display area to show the current input or result.
- Buttons for digits (0–9).
- A decimal point button (.).
- Operation buttons (+, –, *, /).
- An equals button (=).
- A clear button (C).

You do not need to modify the HTML file.



Complete the JavaScript Functionality:

You must complete the logic in **script.js** by implementing the following functions correctly:

1. appendNumber(number)

This function is called when a number button is clicked.
It should:

- Allow multi-digit number input.
- Append the clicked digit to the current input.
- Handle decimal input properly (prevent multiple decimals in a single number).

2. setOperation(operation)

This function is called when an arithmetic operator (+, −, *, /) is clicked.
It should:

- Store the current input as the previous operand.
- Store the selected operation.
- Prepare the calculator to receive the next number.
- Prevent setting an operation if there is no current input.

3. calculate()

This function is called when the equals (=) button is clicked.
It should:

- Perform the selected arithmetic operation using the stored previous value and current value.
- Display the result.
- Handle division by zero by displaying:

`Cannot divide by zero`

If either the previous value or current value is empty, display:

`Error`

4. clearDisplay()

This function is called when the clear (C) button is clicked.
It should:

- Reset the current input.
- Reset the previous input.
- Reset the selected operation.
- Clear the display.

Source Code:

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-
scale=1.0">
    <title>Simple Calculator</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
            background-color: #f4f4f4;
        }
        .calculator {
            border: 1px solid #ccc;
            padding: 20px;
            border-radius: 5px;
            background-color: #fff;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }
        .calculator input, .calculator button {
            margin: 5px;
            padding: 10px;
            font-size: 16px;
        }
        .calculator input {
            width: 100%;
            box-sizing: border-box;
        }
        .calculator button {
            width: 22%;
        }
        .calculator button.operator {
            background-color: #f39c12;
            color: #fff;
        }
    </style>
</head>
<body>
    <div class="calculator">
        <input type="text" id="display" disabled>
```

```
        <br>
        <button onclick="appendNumber('7')">7</button>
        <button onclick="appendNumber('8')">8</button>
        <button onclick="appendNumber('9')">9</button>
        <br>
        <button onclick="appendNumber('4')">4</button>
        <button onclick="appendNumber('5')">5</button>
        <button onclick="appendNumber('6')">6</button>
        <br>
        <button onclick="appendNumber('1')">1</button>
        <button onclick="appendNumber('2')">2</button>
        <button onclick="appendNumber('3')">3</button>
        <br>
        <button onclick="appendNumber('0')">0</button>
        <button onclick="appendNumber('.')">.</button>
        <button onclick="clearDisplay()">C</button>
        <button class="operator" onclick="setOperation('+')">+
</button>
        <button class="operator" onclick="setOperation('-')">-
</button>
        <button class="operator" onclick="setOperation('*')">*
</button>
        <button class="operator"
onclick="setOperation('/')">/</button>
        <br>
        <button style="width: 100%;" onclick="calculate()">=
</button>
    </div>

    <script src="script.js"></script>
</body>
</html>
```

script.js

```
let currentInput = '';
let previousInput = '';
let operation = null;

function appendNumber(number) {
        if (number === '.' && currentInput.includes('.')){
                return;
        }

        currentInput += number;
        updateDisplay();
}

function setOperation(op) {
        if (currentInput === ''){
                return;
        }

        previousInput = currentInput;
        operation = op;
        currentInput = '';
}

function clearDisplay() {
        currentInput = '';
        previousInput = '';
        operation = null;
        updateDisplay();
}

function calculate() {
        if (previousInput === '' || currentInput === ''){
                printToDisplay('Error');
                return;
        }
        let previous = parseFloat(previousInput);
        let current = parseFloat(currentInput);
        let result;

        switch(operation){
                case '+':
                        result = previous + current;
                        break;
                case '-':
```

```
                        result = previous - current;
                        break;
                case '*':
                        result = previous * current;
                        break;
                case '/':
                        if (current === 0){
                                printToDisplay('Cannot divide by
zero');
                                return;
                        }
                        result = previous / current;
                        break;
                default:
                        printToDisplay('Error');
                        return;
        }

        currentInput = result.toString();
        previousInput = '';
        operation = null;
        updateDisplay();
}

function updateDisplay() {
    document.getElementById('display').value = currentInput;
}

function printToDisplay(message) {
    document.getElementById('display').value = message;
}
```